Module 3

Measuring Corporate Performance

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

How Do We Evaluate a Corporation's Performance?

- Who needs to evaluate corporate performance?
  - Managers
  - Current and prospective equity investors
  - Current and prospective debt investors

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- Who needs to evaluate corporate performance?
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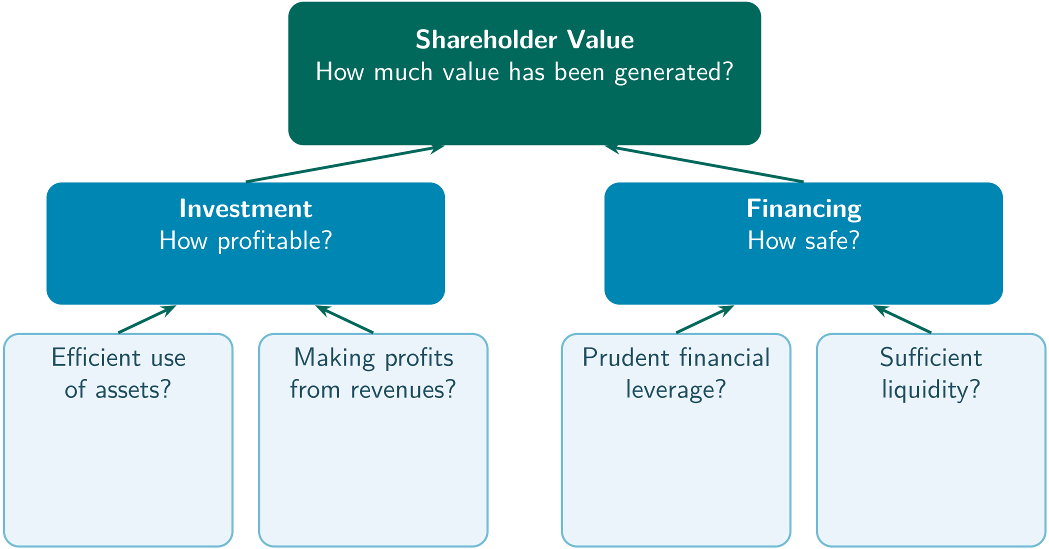
- Corporate performance is **multi-dimensional**

Measuring Corporate Performance



Measuring Corporate Performance

Shareholder Value
How much value has been generated?
Market-to-book ratio

Investment
How profitable?

- Efficient use of assets?
- Making profits from revenues?

Financing
How safe?

- Prudent financial leverage?
- Sufficient liquidity?

Shareholder Value

- Recall from **Module 2**, Market value of equity $=$ Market price of each share $\times$ Total shares outstanding

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- **Market capitalization**, another name for the market value of equity

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- At the end of 2025, share price about \$271, shares outstanding 14,800 Mn

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Example, Apple

- Recall from **Module 2**, Apple's book value of equity at the end of fiscal year 2025 was \$73,733 Mn
- At the end of 2025, share price about \$271, shares outstanding 14,800 Mn
- Market capitalization $=$ \$271 $\times$ 14,800 Mn $=$ \$4,010,800 Mn

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Shareholder Value, Market-to-Book Ratio

- **Market-to-book ratio**, the ratio of the market value of equity to the book value of equity

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Pros

- Closely tied to how actual investors value the company

Cons

- Subject to market fluctuations

Practice Problem

- Here is a simplified balance sheet for a business (figures in $ millions). It has 600 million shares outstanding with a market price of $80 a share.

Current assets	$40,000	Current liabilities	$30,000
Fixed assets	50,000	Long-term liabilities	44,000
		Shareholders' equity	16,000
Total assets	**$90,000**	**Total liabilities and equity**	**$90,000**

- Calculate the market-to-book ratio.

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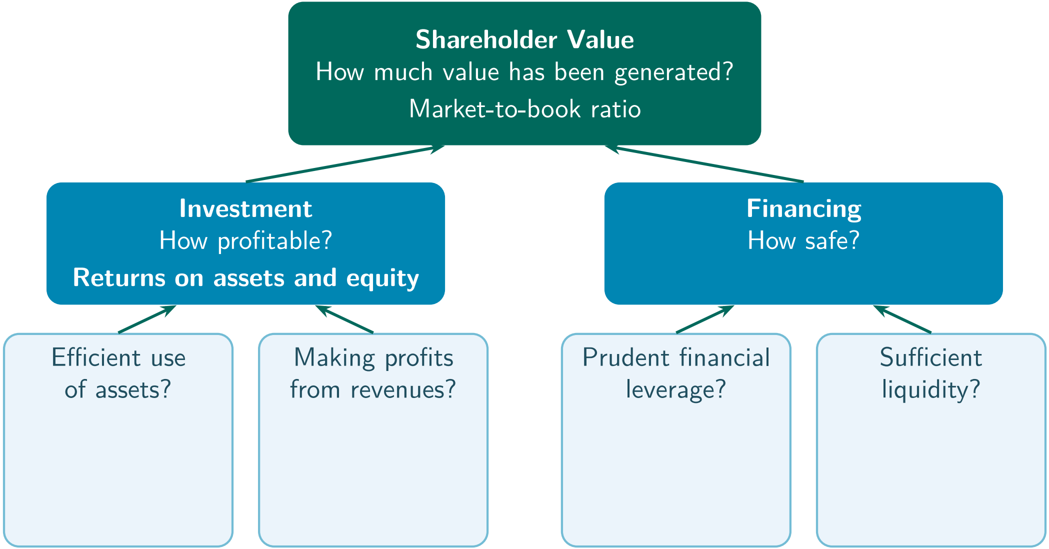
- Calculate the market-to-book ratio. $600 \times \$80 / \$16{,}000 = \$48{,}000 / \$16{,}000 = 3.0$

Measuring Corporate Performance



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- How much profit is generated by each dollar of assets (of equity)?

Pros 👍

- Less sensitive to stock market fluctuations

Cons 👎

- Book values may deviate from market values
- One period, market-to-book reflects all future profits

Example, Income Statement ($ Millions)

	$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	**5,000**
Interest expense	1,000
Taxable income	**4,000**
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Net income	**3,160**
Allocation of net income	
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Example, Balance Sheet ($ Millions)

Assets		Liabilities and equity	
Current assets		Current liabilities	
Cash and marketable sec.	2,600	Debt due for repayment	200
Accounts receivable	500	Accounts payable	9,900
Inventories	9,000	Total current liabilities	10,100
Total current assets	12,100	Long-term liabilities	11,300
Fixed assets		Total liabilities	21,400
Tangible fixed assets			
Property, plant, equipment	48,200	Shareholders' equity	
Less accum. depreciation	19,700	Paid-in capital	6,300
Net tangible fixed assets	28,500	Retained earnings	13,600
Intangible asset	700	Total shareholders' equity	19,900
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Total assets	**41,300**	**Total liabilities and equity**	**41,300**

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$$\text{Return on equity} = \frac{\text{Net income}}{\text{Shareholders' equity}} = \frac{3{,}160}{19{,}900} = 0.159\ (15.9\%)$$

Investment, ROA vs. ROE

$$\text{ROE} = \frac{\text{Net income}}{\text{Equity}} = \frac{\text{Net income}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}} = \text{ROA} \times \frac{\text{Assets}}{\text{Equity}}$$

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  - Borrowing money makes ROE higher than ROA
  - As we see in **Leverage**, borrowing money makes ROE more volatile than ROA

Practice Problem

- Last year's ROE was 30%. This year the ROE has decreased to 20%, even though the firm's net income equaled last year's. What caused the decrease?

a. Shareholders' equity decreased by 10%.
b. Shareholders' equity decreased by 50%.
c. Shareholders' equity increased by 10%.
d. Shareholders' equity increased by 50%.

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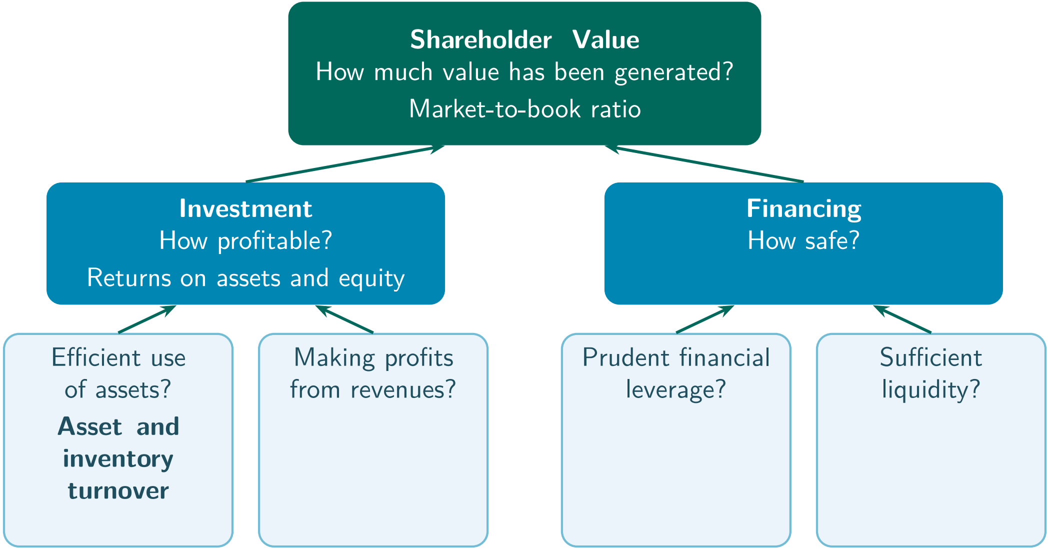
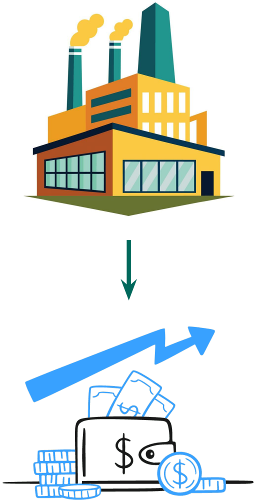
- Suppose net income is X in both years. Then $30\% = X$ / shareholders' equity last year, and $20\% = X$ / shareholders' equity this year
- Then $X = 30\% \times$ shareholders' equity last year $= 20\% \times$ shareholders' equity this year
- So shareholders' equity this year / shareholders' equity last year $= 30\%/20\%$ $= 150\%$, an increase of 50%

Measuring Corporate Performance



Efficiency, Use of Assets

$$\text{Asset turnover ratio} = \frac{\text{Revenues}}{\text{Total assets}}$$



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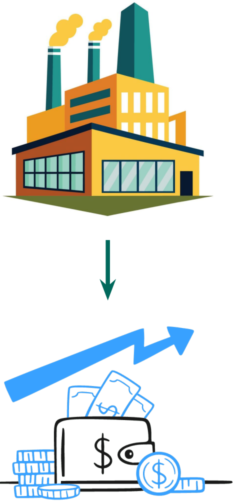
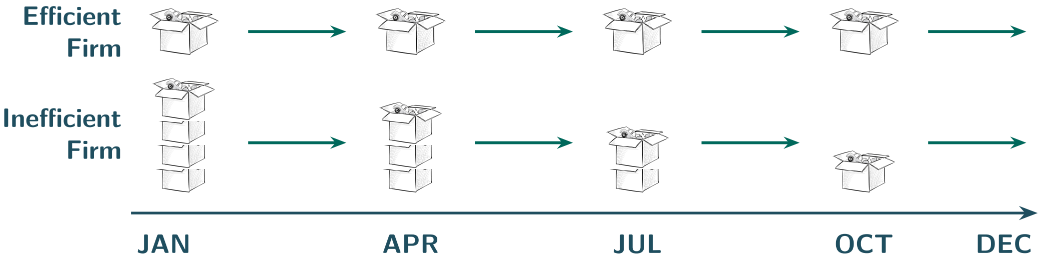


Efficiency, Inventory



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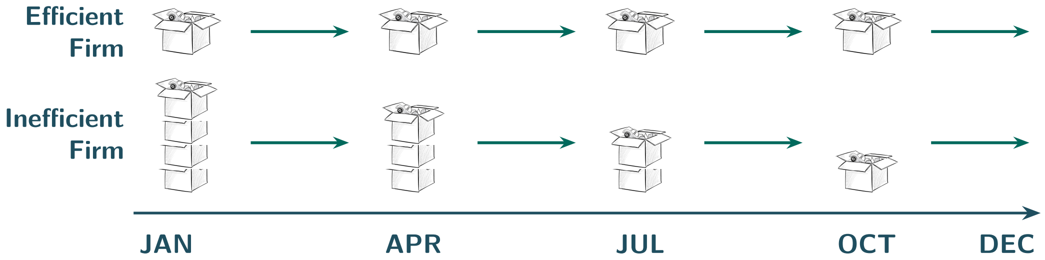
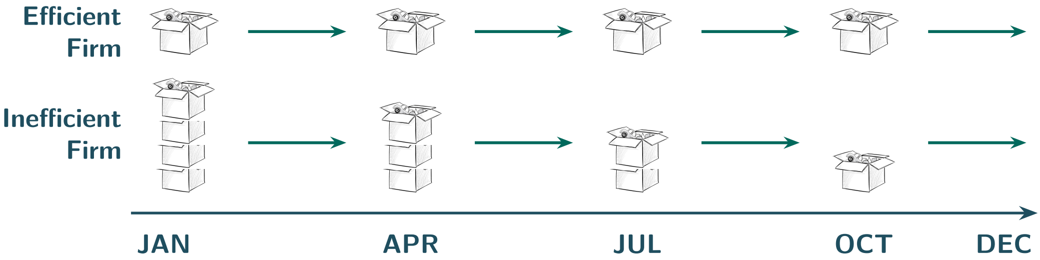


$$\text{Inventory turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$$

Efficiency, Inventory



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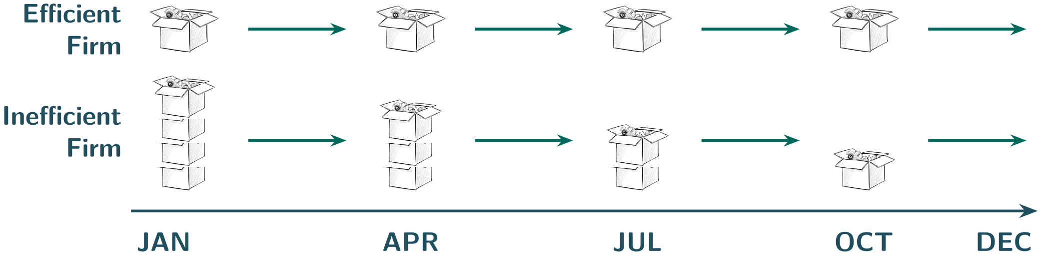

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- Does the company turn over inventories rapidly?
- Efficient firms do not let inventories sit idle
- Aim for the highest possible? No, inadequate inventories may fail to meet customer demand in time

Efficiency, Accounts Receivable

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- The average length of time for customers to pay their bills
- Efficient firms collect bills quickly
- Aim for the fastest possible? No, it may reflect an over-stringent credit policy
  - **Credit policy**, the terms on which a firm lets customers buy now and pay later

Example, Income Statement ($ Millions)

	$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	**5,000**
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Allocation of net income	
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Example, Balance Sheet ($ Millions)

Assets		Liabilities and equity	
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Example, Efficiency Ratios

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$$\text{Inventory turnover ratio} = \frac{56{,}000}{9{,}000} = 6.2$$

$$\text{Average collection period} = \frac{500}{80{,}000/365} = 2.3 \text{ days}$$

Practice Problem

- A firm had accounts receivable of \$940 and generated \$3,400 of revenues during the year. How much is its average collection period?

a. 94 days

b. 95 days

c. 101 days

d. 107 days

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d. 107 days

- Average collection period = 940 / (3,400 / 365) = 101 days

Measuring Corporate Performance

Shareholder Value
How much value has been generated?
Market-to-book ratio

Investment
How profitable?
Returns on assets and equity

Efficient use of assets?
Asset and inventory turnover

Making profits from revenues?
Profit margin

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- How much of each dollar of revenues ends up as net income?

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Example, Profit Margin

$$\text{Profit margin} = \frac{3{,}160}{80{,}000} = 0.0395\ (3.95\%)$$

Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?

Aswath Damodaran, "Dean of Valuation"

Excess Returns, January 22, 2026 (1:23), "Could AI Actually Reduce Profit Margins? Aswath Damodaran Thinks So"

Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?
- Do you agree?

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Decompose ROA

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Decompose ROA

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$$\text{ROA} = \text{Asset turnover} \times \text{Profit margin}$$

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$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}} = \text{Asset turnover} \times \text{Profit margin} = 1.94 \times 0.0395 = 0.077$$

Decompose ROA

Company	Strategy	Asset Turnover	Profit Margin	ROA
Zara	Fast fashion	1.6	7.5%	12%
Hermès	Luxury fashion	0.55	22.5%	12%

Practice Problem

- Which of the following will allow your firm to achieve its targeted 16% ROA with an asset turnover of 2.5?

a. A current ratio of 1.6.
b. A P/E ratio of 14.
c. A return on equity of 25%.
d. A profit margin of 6.4%.

Practice Problem

- Which of the following will allow your firm to achieve its targeted 16% ROA with an asset turnover of 2.5?

a. A current ratio of 1.6.
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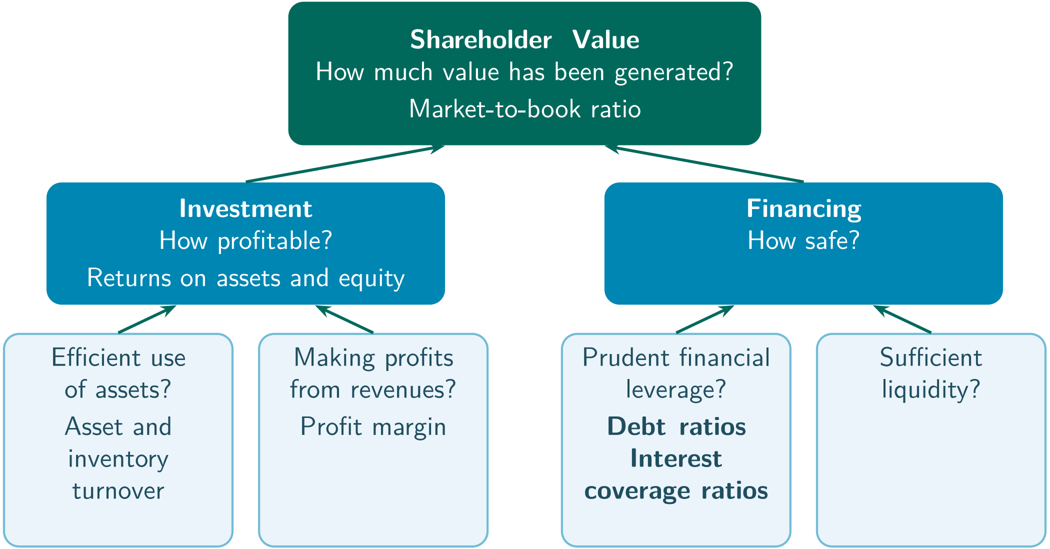
- ROA $=$ Asset turnover $\times$ Profit margin, and $2.5 \times 6.4\% = 16\%$

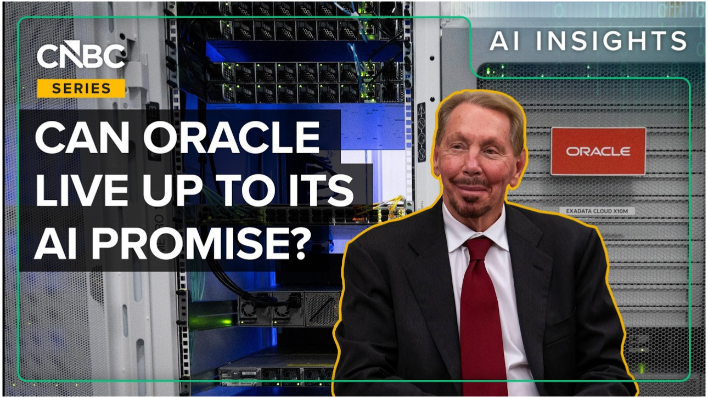
Recap, Investment

- Investment profitability is measured with the returns on assets and equity
- Efficiency ratios, how much in revenues the assets generate, how quickly inventories sell, and how quickly customers pay
- Profit margin, how much of each dollar of revenues ends up as net income
- ROA $=$ Asset turnover $\times$ Profit margin

Measuring Corporate Performance



Motivating Example, Oracle



CNBC, March 10, 2026 (5:45), "How Oracle's AI-Fueled Debt Load Has Investors On Edge"

Example, Financing a Data Center

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All equity financed

Good
\$6B (200%)

Bad
\$2B (0%)

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Equity holders	\$4.5B (350%)
Debt holders	\$1.5B

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Debt holders	\$1.5B

Bad

Equity holders	\$0.5B (−50%)

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

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\$1B equity + \$1B debt

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	Good		**Bad**
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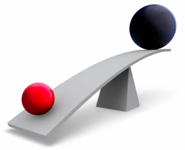
Debt holders	\$1.5B	Debt holders	\$1.5B

With debt, gains and losses to equity holders are amplified

Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand



Financial Leverage

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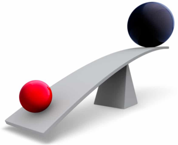
- Expansion succeeds and profits rise $\rightarrow$ debt holders still receive only the fixed interest and principal, the gains go to shareholders



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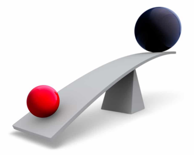

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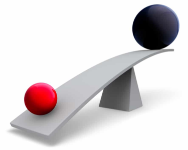

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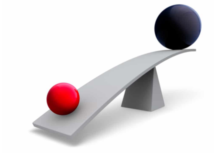
  - This is **financial leverage**



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- The debt-to-equity ratio is also called the **leverage ratio**
  - It tells to what extent returns to equity investors are amplified because of debt
  - Higher leverage ratio = higher risk of default
    - **Default**, the firm fails to make a required debt payment
    - Default can lead to **bankruptcy**

Example, Balance Sheet ($ Millions)

Assets		**Liabilities and equity**	
Current assets		Current liabilities	
Cash and marketable sec.	2,600	Debt due for repayment	200
Accounts receivable	500	Accounts payable	9,900
Inventories	9,000	Total current liabilities	10,100
Total current assets	12,100	Long-term liabilities	11,300
Fixed assets		Total liabilities	21,400
Tangible fixed assets			
Property, plant, equipment	48,200	Shareholders' equity	
Less accum. depreciation	19,700	Paid-in capital	6,300
Net tangible fixed assets	28,500	Retained earnings	13,600
Intangible asset	700	Total shareholders' equity	19,900
Total fixed assets	29,200		
Total assets	**41,300**	**Total liabilities and equity**	**41,300**

Example, Leverage

$$\text{Debt-to-equity ratio} = \frac{\text{Total liabilities}}{\text{Shareholders' equity}} = \frac{21{,}400}{19{,}900} = 1.08$$

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- Ability of the firm to pay the interest on its debt

Example, Income Statement ($ Millions)

	$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	**5,000**
Interest expense	1,000
Taxable income	**4,000**
Taxes	840
Net income	**3,160**
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Example, Leverage

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest expense}} = \frac{5{,}000}{1{,}000} = 5.0$$

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$$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest expense}} = \frac{5{,}000 + 3{,}000}{1{,}000} = 8.0$$

Practice Problem

- A firm pays an 8% rate of interest on $10 million of outstanding debt with face value $10 million. The firm's EBIT was $1 million.

a. What is its times interest earned?

b. If depreciation is $200,000, what is its cash coverage ratio?

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   Interest expense $= 0.08 \times$ \$10 million $=$ \$800,000
   Times interest earned $=$ \$1,000,000 / \$800,000 $= 1.25$
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   Times interest earned $=$ \$1,000,000 / \$800,000 $=$ 1.25
b. If depreciation is \$200,000, what is its cash coverage ratio?
   Cash coverage ratio $=$ (\$1,000,000 $+$ \$200,000) / \$800,000 $=$ 1.5

Practice Problem

- Which one of the following will increase a firm's times interest earned ratio?

a. An increase in debt.
b. A decrease in cost of goods sold.
c. An increase in interest expense.
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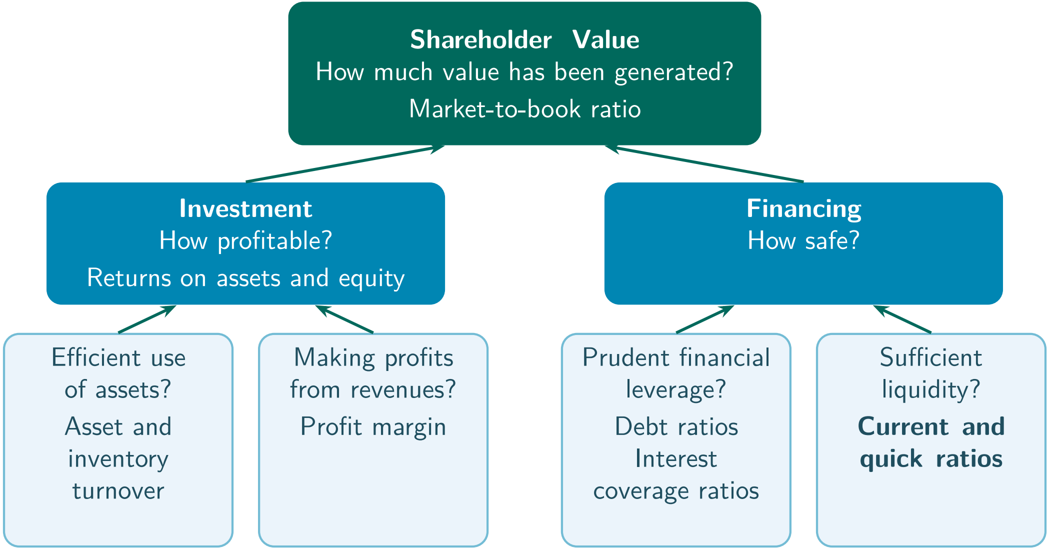
- Lower COGS raises EBIT, the numerator

Measuring Corporate Performance



Financial Liquidity

- **Liquidity**, the ability to sell an asset on short notice at close to the market value

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- Liquid assets can be converted into cash quickly and cheaply

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Most liquid → Least liquid

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Paid first → Paid last

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- Liquidity metrics compare the amount of liquid assets to the amount of debt that requires immediate repayment

Example, Liquidity

$$\frac{\text{Current assets}}{\text{Current liabilities}} = \frac{12{,}100}{10{,}100} = 1.20$$

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$$\frac{\text{Cash} + \text{Marketable securities} + \text{Accounts receivable}}{\text{Current liabilities}} = \frac{2{,}600 + 500}{10{,}100} = 0.31$$

Practice Problem

- Consider this simplified balance sheet for a business.

Balance Sheet ($)			
Current assets	$100	Current liabilities	$60
Long-term assets	500	Long-term debt	280
		Other liabilities	70
		Equity	190
Total	**$600**	**Total**	**$600**

a. What is the company's debt-to-equity ratio?

b. What is its current ratio?

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b. What is its current ratio? $\$100/\$60 = 1.67$

Practice Problem

- A firm uses $1 million in cash to purchase inventories.

a. Will its current ratio rise or fall?

b. Will its quick ratio rise or fall?

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a. Will its current ratio rise or fall?
   The current ratio is unaffected. Inventories replace cash, but total current assets are unchanged.

b. Will its quick ratio rise or fall?
   The quick ratio falls, since inventories are not included in the most liquid assets.

Recap, Financing

- Financial leverage, debt raises shareholder returns in good times and lowers them in bad times
- Debt-to-equity ratio (the leverage ratio) and debt ratio, how much debt the firm carries
- Times interest earned and cash coverage ratio, the ability to pay the interest on its debt
- Current and quick ratios, do liquid assets cover the debt that requires immediate repayment?

Key Takeaways

- Shareholder value, the market-to-book ratio

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- Investment, the returns on assets and equity, efficiency ratios, and the profit margin

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- Investment, the returns on assets and equity, efficiency ratios, and the profit margin
- Financing, financial leverage, debt and coverage ratios, and liquidity ratios
- Corporate performance is multi-dimensional, no single ratio captures it

Thank you!

Please let me know if you have any questions.